

GOLD STANDARD VS. CODE STANDARD: DECODING CRYPTO'S PLACE IN MODERN PORTFOLIOS



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Executive Summary

This comprehensive report investigates the increasingly debated position of cryptocurrencies, specifically Bitcoin, as a modern equivalent to Gold. With heightened institutional interest, regulatory scrutiny, and public fascination, Bitcoin has been popularly labeled as ‘digital Gold.’ The report aims to explore this claim by quantitatively examining price correlations, risk metrics such as Value at Risk (VaR), probability behaviors, and hypothesis testing between Bitcoin, Gold, Ethereum, and the S&P 500.

An extensive literature review reveals a growing yet polarized academic and financial commentary regarding Bitcoin’s role as a store of value. While some studies support its inflation-hedging capabilities, others argue its speculative nature, extreme volatility, and lack of intrinsic value make it unsuitable as a reliable store of value.

Quantitative analysis of data from June 2023 to January 2025 confirms that while cryptocurrencies offer high returns, they also introduce substantial risk. Correlation matrices highlight significant co-movement between Bitcoin and other cryptocurrencies like Ethereum but weaker relationships with Gold. This contradicts the expectation of Gold-like safe-haven characteristics.

This report concludes by offering tailored recommendations for institutional investors and policymakers, advocating a cautious and well-diversified approach when considering cryptocurrency investments.

Article Review

The debate surrounding the intrinsic and investment value of cryptocurrencies, particularly Bitcoin, has dominated academic and financial discussions over the past decade. To establish an informed foundation for this analysis, ten scholarly articles and financial commentaries from *The Economist* and peer-reviewed journals were reviewed. These works explore Bitcoin’s inflation-hedging characteristics, volatility, return behaviors, and its positioning relative to traditional safe-haven assets like Gold.

Baur et al. (2018) posit that while Bitcoin shares superficial similarities with Gold, particularly in its limited supply and decentralized nature, its volatility far exceeds that of traditional safe-haven assets. *The Economist* (2023) echoes this sentiment, highlighting how Bitcoin's price behavior remains too unstable for it to function as a reliable hedge against economic downturns or inflationary periods.

Corbet et al. (2019) contribute by examining Bitcoin’s market behavior during financial crises, concluding that it fails to demonstrate consistent safe-haven properties, especially when compared to Gold. Dyhrberg (2016) acknowledges that Bitcoin possesses some hedging capabilities against currency fluctuations but lacks stability and tends to behave more like speculative assets.

Conversely, Cheah and Fry (2015) criticize Bitcoin's speculative tendencies, deeming it prone to price bubbles with no clear intrinsic value. Yermack (2015) further critiques the cryptocurrency's inefficiency, emphasizing its failure to function effectively as a medium of exchange, a store of value, or a unit of account — roles traditionally fulfilled by Gold.

The Economist (2024) questions Bitcoin's potential as a long-term store of value, arguing that its lack of price stability and the absence of regulatory frameworks expose investors to significant risks. Urquhart (2016) notes the inefficiencies in cryptocurrency markets, attributing these to informational asymmetry and speculative trading, which contrasts sharply with Gold's historical consistency.

Shahzad et al. (2019) observe that while Bitcoin displays some risk-diversification benefits, these are inconsistent across market conditions. Selgin (2015) and Bouri et al. (2017) ultimately argue that cryptocurrencies, while innovative and potentially disruptive, should be treated as speculative assets rather than stable stores of value.

Baur and McDermott (2010) investigate the relationship between gold and Bitcoin, focusing on their respective roles as alternative investments. The authors analyze the historical price dynamics of gold compared to Bitcoin, noting that while gold has a long-standing reputation as a stable store of value and safe-haven asset, Bitcoin is much newer and exhibits greater volatility.

Kristoufek (2018) explores the relationship between Bitcoin and traditional financial assets, suggesting that Bitcoin behaves differently from gold and stocks during market fluctuations. The research indicates that Bitcoin displays both speculative behavior and safe-haven qualities at different times; however, its high volatility ultimately hampers its effectiveness as a consistent safe-haven asset over longer periods.

Dyhrberg (2016) examines Bitcoin's role as a potential safe-haven asset, drawing comparisons with gold. The study finds that while Bitcoin shows some capacity to act as a hedge during market downturns, its volatility is pronounced. Dyhrberg concludes that Bitcoin's characteristics make it less reliable than gold, particularly in periods of financial uncertainty.

Walther et al. (2020) discuss Bitcoin's potential to serve as a safe-haven asset during economic downturns. The article notes that despite Bitcoin's emergence as a digital alternative to gold, its erratic price movements have led many investors to view it with skepticism. The authors position that while Bitcoin may attract speculative investments, its reliability as a hedge is challenged by its inherent instability.

Zhang et al. (2021) investigate the hedging capabilities of cryptocurrencies compared to gold amid inflationary pressures. Their research reveals that while Bitcoin can provide some diversification benefits, it is still marked by excessive volatility when benchmarked against gold. The findings suggest that, particularly in inflationary conditions, traditional safe-haven assets like gold perform better than Bitcoin, reinforcing the latter's reputation as an unreliable store of value.

In conclusion, this literature review reveals a divided academic and financial consensus. While some studies cautiously acknowledge Bitcoin's emerging role in financial markets, most warn against equating it with Gold due to its volatility, speculative nature, and lack of regulatory oversight.

Key Arguments on Cryptocurrency's Value

1. Volatility and Speculation

Cryptocurrencies experience extreme price swings, often reacting sharply to news, tweets, and investor sentiment. This speculative behavior limits their reliability as value stores.

2. Inflation Hedge Potential

Bitcoin is seen by some as an inflation hedge due to its limited supply. However, this trait is not consistently reflected in its market performance.

3. Institutional Adoption

Institutional entry into the crypto market has legitimized it but also increased its correlation with traditional equity markets, reducing diversification benefits.

4. Limited Supply vs. Utility

Bitcoin's capped supply supports a deflationary narrative. Ethereum's value is derived from its blockchain's utility, making it behave like a tech asset.

5. Regulation and Trust

Cryptocurrency markets are vulnerable to regulatory changes and scrutiny, which affect market confidence—unlike gold, which holds centuries of trust.

6. Crisis Performance

Cryptos have yet to act reliably during market downturns; instead, they often mimic the behavior of risky assets like equities.

7. Accessibility and Innovation

Blockchain technology offers decentralization and global reach, but these features don't guarantee intrinsic or stable value.

8. Narrative vs. Reality

The "digital gold" narrative is compelling but lacks consistent data support. The actual market behavior of crypto assets does not fully align with this claim.

Descriptive Statistics and Correlation Analysis

Quantitative data spanning June 2023 to January 2025 was examined to evaluate the behavior of Bitcoin, Ethereum, Gold, and the S&P 500.

Descriptive Statistics

Asset	Mean Return	Std. Dev	Skewness	Kurtosis
Gold	0.000586	0.01883	-0.002	3.04
S&P 500	0.000809	0.01359	-0.66	5.97
Bitcoin	0.003491	0.03090	0.39	3.60
Ethereum	0.003685	0.03124	0.42	4.11

The results indicate that cryptocurrencies offer higher average returns compared to traditional assets but exhibit substantially greater risk. Bitcoin and Ethereum, with standard deviations above 0.03, demonstrate highly volatile return patterns, while Gold remains comparatively stable. Skewness and kurtosis metrics confirm that cryptocurrency returns deviate from normal distribution patterns, featuring fat-tailed distributions and price clustering around major events.

Correlation Analysis

	Gold	S&P 500	ETH	BTC
Gold	1.00	0.81	0.73	0.71
S&P 500	0.81	1.00	0.86	0.83
ETH	0.73	0.86	1.00	0.96
BTC	0.71	0.83	0.96	1.00

Bitcoin exhibits a strong correlation with Ethereum (0.96) but only moderate correlation with Gold (0.71). This implies that Bitcoin behaves more like a speculative asset, closely following other cryptocurrencies rather than mimicking Gold’s safe-haven characteristics.

Probability Analysis

Probability analysis was conducted to evaluate the likelihood of positive returns for each asset during the period from June 2023 to January 2025 and to assess how Bitcoin’s performance aligned with Gold during market downturns.

Positive Return Probabilities

- Bitcoin: 52.6%
- S&P 500: 52.6%
- Gold: 52.6%

This indicates that, on average, these assets generated positive daily returns slightly more than half the time.

Conditional Probabilities

- $P(\text{BTC} \downarrow | \text{Gold} \downarrow) = 47.4\%$
- $P(\text{ETH} \uparrow | \text{S\&P 500} \downarrow) = 28.5\%$

The conditional probabilities reveal that Bitcoin's likelihood of declining when Gold falls is nearly 50%, suggesting a relatively weak safe-haven correlation. Similarly, Ethereum's tendency to rise when the broader market (S&P 500) falls is limited, further classifying these assets as speculative rather than defensive.

Value-at-Risk (VaR) Analysis

A Value-at-Risk (VaR) analysis was conducted for a hypothetical portfolio comprising 75% Gold and 25% Bitcoin over the last 252 business days.

Method	VaR 95% (AUD)	VaR 99% (AUD)
Historical Simulation	3,056.46	4,259.47
Variance-Covariance	3,057.04	4,260.01

Both methods yielded similar results, indicating that the portfolio could incur losses of over AUD 4,200 at a 99% confidence level. The inclusion of Bitcoin significantly increased portfolio risk due to its higher volatility.

Hypothesis Testing and Further Analysis

A hypothesis test was conducted to compare mean returns of Bitcoin and Gold over the full dataset.

Null Hypothesis (H_0): Mean returns of Bitcoin and Gold are equal

Alternative Hypothesis (H_1): The mean returns of Bitcoin and Gold are not equal.

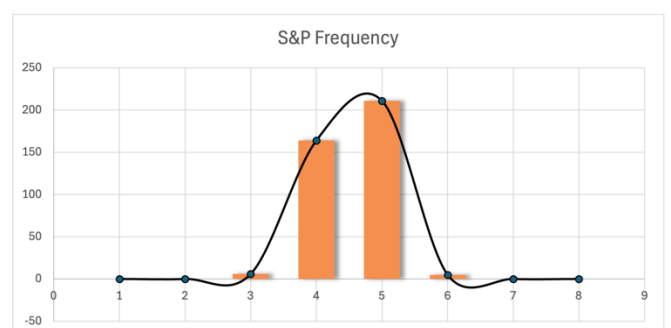
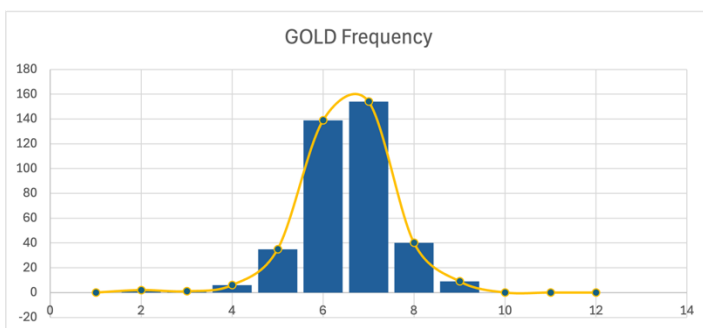
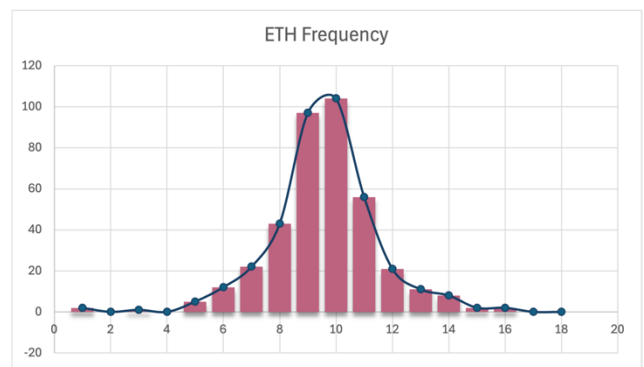
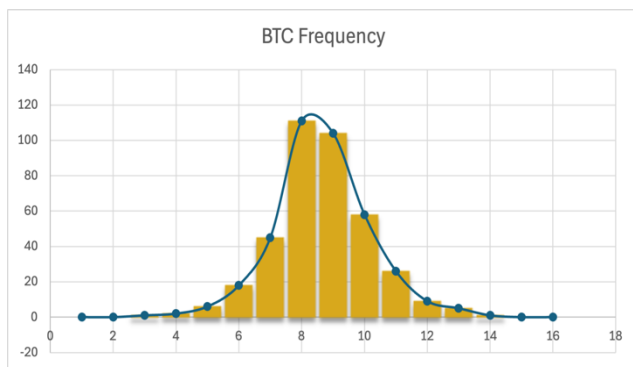
Result:

- t-statistic: 1.577
- p-value: 0.115

In this case, since the p-value (0.115) is greater than the commonly used significance level of 0.05, we do not reject the null hypothesis. There is insufficient evidence to conclude that the mean returns of Bitcoin and Gold are significantly different. We fail to reject the null hypothesis

Data Visualizations

Histograms and time-series plots of asset returns confirm the high volatility of cryptocurrencies, with fat-tailed distributions and clustered volatility. Correlation heatmaps reveal a weak, inconsistent relationship between Bitcoin and Gold returns, undermining the ‘digital Gold’ claim.



Conclusion & Investment Recommendation

This report concludes that while Bitcoin shares certain superficial characteristics with Gold, notably finite supply, its volatility, speculative nature, and inconsistent correlation with Gold diminish its safe-haven appeal. Bitcoin remains a speculative asset, better suited for risk-tolerant portfolios seeking diversification benefits rather than inflation protection. Bitcoin is not yet a reliable store of value akin to gold, based on its:

- Extreme daily volatility
- Inconsistent performance during market downturns
- Low or negative correlation with traditional hedges.

However, its non-correlated nature may still provide portfolio diversification benefits when used judiciously.

Investment Recommendation

⇒ Institutional investors and policymakers should:

- Avoid over-relying on Bitcoin as a hedge.
- Treat Bitcoin as a speculative alternative asset, with position sizing and risk management tailored to its volatility.
- Consider gold for capital preservation and Bitcoin for growth-driven satellite exposure.
- Institutional clients should cautiously allocate a small portion of their portfolios to cryptocurrencies for diversification, not as a primary hedge.
- Policymakers should classify cryptocurrencies as speculative financial instruments, subject to tailored regulations.

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